

# **Toward a Formal Ontology of Created Reality**

## **(Tegmark-Aquinas Conceptual Ecosystem)**

### **Abstract**

This monograph develops a unified ontological framework for understanding the relationship between quantum neurodynamics, biological organization, and conscious experience within the Theory of Actualized Conscious Experience (TACE). Rather than proposing consciousness as an emergent property of physical complexity, the work situates quantum physics, neuroscience, phenomenology, biology, mathematics, and metaphysics within four integrated levels of one coherent ontology: the Semantic–Ontological, Formal–Mathematical, Physical, and Phenomenological–Biological Levels.

Within this framework, the Matrix constitutes the universal formal order governing every lawful actualization of creation, while the Q-Form embodies the structured physical potentiality through which quantum systems become historically realized. Quantum field theory and the wave function describe the lawful physical evolution of these structured possibilities, whereas physical interactions continuously transform realized states through successive actualizations. Consequently, quantum mechanics is interpreted not as the deepest level of reality but as the physical description of one level within a broader ontological architecture.

The neuropsychic organization of the brain is understood as the embodied physical realization of this lawful order. Global quantum configurations, mathematically representable by tensor product states, provide instantaneous descriptions of the physical organization of the nervous system. Macroscopic electromagnetic activity and biological perception emerge as statistically integrated manifestations of countless microscopic physical actualizations occurring throughout the neuropsychic substrate. These processes explain the physical embodiment of cognition but do not by themselves account for conscious understanding.

Following the biological insights of Humberto Maturana and Francisco Varela, TACE recognizes autopoiesis and structural coupling as the organizational principles governing living systems and operational cognition. These principles explain biological embodiment and adaptive interaction with the environment but remain insufficient to explain intentional understanding, universal concepts, truth, or self-reflective consciousness.

Within TACE, the human person is understood as one unified substance composed of body and substantial rational soul. Consciousness is not produced by quantum fields, recursive computation, or biological complexity, but belongs to the proper activity of the substantial soul acting through the powers of intellect, intentionality, and will. The lawful correspondence between neuropsychic realization and intentional understanding is mediated by the Silver Bridge, the universal onto-epistemological principle through which the Physical Realm and the Realm of Forms remain intrinsically interconnected.

This ontological architecture provides a reformulation of the classical mind-brain problem. Rather than attempting to reduce consciousness to physical processes or separating mind and matter into independent substances, TACE demonstrates that both positions arise from an incomplete ontology. The brain belongs to the Physical Level as the embodied historical realization of the person, while conscious understanding belongs to the Semantic–Ontological Level through the activity of the substantial soul. Their lawful coordination is established through the Silver Bridge within the universal order governed by the Matrix and physically realized through the Q-Form.

Accordingly, TACE proposes not merely a theory of consciousness but a comprehensive formal ontology of created reality. By integrating Aristotelian-Thomistic metaphysics, modern mathematics, quantum physics, neuroscience, the biological theory of autopoiesis, and phenomenology into a single coherent framework, it provides a unified account of how mathematical order becomes physically realized, biologically embodied, intentionally understood, and consciously experienced. In this way, the classical mind-brain problem is not solved through reduction or dualistic separation, but dissolved by recognizing consciousness and embodiment as complementary expressions of one unified ontological order grounded in the Creator.

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# Chapter I. The Formal and Physical Ground of Neurodynamics

To understand the physical substrate of neurodynamics, one must move beyond the classical conception of matter as a collection of isolated particles and adopt the modern field-theoretic description provided by Quantum Field Theory (QFT). Within TACE, however, quantum field theory belongs to the **Physical Level** of description and must itself be understood within a broader ontological framework.

The **Formal–Mathematical Level** of TACE is constituted by the Matrix, the universal formal order governing the admissible actualizations of creation. The Matrix does not consist of physical entities, fields, or particles; rather, it establishes the lawful mathematical structure according to which physical actualizations may occur. Within the Physical Level, this formal order is realized through the **Q-Form**, the ontological principle of structured physical potentiality.

Quantum fields therefore do not constitute the ultimate ontological reality, but the fundamental physical substrate through which the Matrix-governed Q-Form becomes physically realized. The quantum vacuum, mathematically represented by the ground state  $|0\rangle$ , describes the lowest-energy configuration of the physical field rather than the ultimate foundation of being itself. It belongs entirely to the Physical Level and must be distinguished from the ontological principles that govern its lawful behavior.

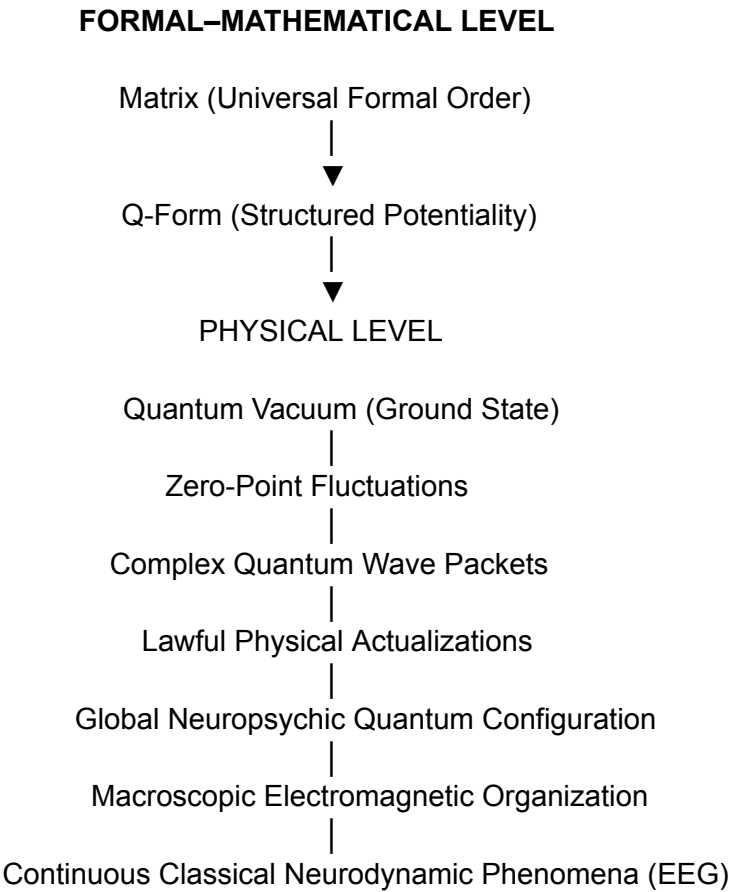
Although the quantum vacuum possesses minimal energy, it is not physically inert. Consistent with quantum field theory, zero-point fluctuations continuously arise within the physical field according to the uncertainty relations governing quantum systems. These fluctuations are not violations of physical law but lawful manifestations of the structured potentiality embodied by the Q-Form.

Within TACE, isolated monochromatic waves are regarded as mathematical idealizations. Physical wave propagation always occurs as complex wave packets composed of multiple interacting frequencies and amplitudes. These wave packets describe the mathematical evolution of quantum states and the corresponding structured physical possibilities available to a system.

As the amplitudes within a physical wave packet approach the smallest physically meaningful magnitudes, they encounter the limits imposed by the quantum structure of spacetime and the Planck scale. At these scales, the continuous mathematical description of the wave corresponds physically to discrete quantum actualizations. These discrete quantum states should not be interpreted as isolated particles detached from continuous reality, but as localized physical realizations within the lawful space of admissible actualizations governed by the Matrix through the Q-Form.

Within the neuropsychic organization of the brain, immense populations of quantum states coexist and interact continuously. At every moment these microscopic physical realizations collectively determine the instantaneous quantum configuration of the neuropsychic system. The resulting global physical state provides the physical substrate from which macroscopic electromagnetic organization emerges.

Classical neuroscientific instruments cannot resolve the individual quantum realizations composing this microscopic organization. Consequently, electroencephalographic activity, electromagnetic oscillations, and ordinary biological perception reveal statistically integrated macroscopic patterns rather than the underlying quantum discreteness. The apparent continuity of classical neurodynamics therefore reflects the large-scale statistical organization of innumerable microscopic physical actualizations.



The introduction of thermal energy, chemical activity, or biological metabolism modifies the physical state of the quantum fields by generating real excitations above the vacuum state. These processes alter the physical realization of the system without affecting the lawful ontological order established by the Matrix or

the structured potentiality embodied in the Q-Form. The quantum vacuum therefore remains the lowest-energy physical configuration of the field, while the ontological principles governing physical realization remain invariant throughout every lawful process of actualization.



# Chapter II. The Lawful Actualization of Quantum Wave Structures

Within the Physical Level of TACE, isolated monochromatic waves are regarded as mathematical idealizations rather than physically realizable entities. Physical wave propagation occurs through complex wave packets composed of multiple interacting frequencies, amplitudes, and phases, representing the mathematical evolution of quantum states within the lawful structure of physical reality.

The continuous mathematical description of these wave packets does not imply that every physical realization possesses unlimited continuity. Rather, the Matrix governs the admissible physical actualizations through the structured potentiality embodied in the Q-Form. Consequently, the mathematical continuity of quantum wave functions corresponds physically to a finite domain of lawful realizations constrained by the physical structure of spacetime.

## The Physical Limits of Continuous Wave Descriptions

As the amplitudes composing a quantum wave packet diminish toward the smallest physically meaningful magnitudes, they approach the limits imposed by the quantum structure of spacetime. The Planck scale therefore represents a fundamental physical boundary beyond which current physical descriptions cease to possess direct operational meaning.

Within TACE, this boundary should not be understood as the destruction of continuity itself, but as the transition between continuous mathematical description and discrete physical realization. Mathematical continuity remains an ideal property of the formal description, whereas physical actualization necessarily occurs through finite, localized quantum states governed by the Matrix through the Q-Form.

## Structured Potentiality and Quantum Actualization

The Q-Form does not determine a single physical outcome but embodies the lawful space of admissible physical possibilities available to a physical substance at a given moment. Every physical interaction explores this structured potentiality according to the governing constraints established by the Matrix.

When a lawful physical interaction reaches an actualizing condition, one admissible physical realization becomes historically actualized. This transition should not be interpreted as the collapse of mathematical continuity but as the lawful ontological actualization of structured potentiality into a new physical state (**q-form<sub>i</sub>**).

Accordingly, microscopic physical reality should be understood neither as perfectly continuous nor as intrinsically fragmented. Rather, physical systems continuously evolve according to quantum dynamics while becoming historically determinate through successive lawful actualizations.

The quantum wave function therefore describes the mathematical evolution of structured physical possibilities, whereas the Q-Form constitutes the ontological principle governing the admissible realization of those possibilities within the Physical Realm.



# Chapter III. The Global Neuropsychic Quantum Configuration

When applied to neuroscience, the TACE framework provides an ontological interpretation of quantum neurodynamics without reducing consciousness to quantum mechanics. Rather than seeking fragile macroscopic quantum coherence as the source of consciousness, TACE distinguishes between the physical organization of the neuropsychic and the ontological principles governing conscious experience.

Traditional quantum-consciousness theories, such as Penrose and Hameroff's Orchestrated Objective Reduction (Orch-OR), propose that biologically protected quantum coherence within neuronal microtubules is essential for consciousness. TACE adopts a different perspective. The physical organization of the brain is understood as a lawful quantum system whose behavior is governed by the Matrix through the structured potentiality of the Q-Form. Consciousness itself, however, is not generated by quantum coherence but belongs to the activity of the substantial human soul.

Biological heat continuously produces thermal excitations throughout the nervous system. These excitations alter the instantaneous physical configuration of neuronal tissues and rapidly suppress large-scale quantum coherence. Nevertheless, the fundamental physical constants and the lawful quantum structure of matter remain unchanged. Thermal activity modifies only the particular physical realization of the neuropsychic system; it does not alter the Matrix, the Q-Form, or the physical laws governing quantum interactions.

At every instant, the immense number of quantum states distributed throughout the neuropsychic organization forms a single global mathematical description of the physical brain. Formally, this instantaneous configuration may be represented by a tensor product state encompassing the quantum degrees of freedom participating in the neural architecture. This tensor product state constitutes a complete mathematical snapshot of the physical organization of the brain at a particular moment in its historical evolution.

Within TACE, this global tensor product state belongs entirely to the **Physical Level**. It describes the instantaneous quantum organization of the neuropsychic substrate but neither constitutes nor generates consciousness. Instead, it represents the physical condition through which conscious acts may become historically embodied.

Each lawful physical interaction transforms the preceding physical realization ( $q\text{-form}_0$ ) into a new realized state ( $q\text{-form}_1$ ). Consequently, the tensor product state is continuously updated as the physical history of the neuropsychic system unfolds through successive actualizations governed by the Matrix. The mathematical description therefore evolves continuously, while the underlying physical organization acquires an ever richer historical structure through lawful actualization.

The transition from this physical neuropsychic organization to conscious understanding is not explained by the tensor product state itself. Rather, it is mediated by the **Silver Bridge**, the universal onto-epistemological principle through which neuropsychic realizations become intentionally present to the substantial



human soul. Through this lawful mediation, acts of intellect, intentionality, and will become historically embodied without reducing consciousness to the physical processes that condition its expression.

Accordingly, the tensor product state should be understood as the instantaneous mathematical representation of the embodied neuropsychic organization, whereas conscious experience belongs to the proper activity of the substantial soul. Quantum neurodynamics provides the lawful physical substrate of cognition, while the Silver Bridge establishes the correspondence through which physical realization and intentional understanding become integrated within the single ontological framework of TACE.

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# Chapter IV. The Macroscopic Manifestation of Neuropsychic Organization

The transition from quantum neurodynamics to classical neurophysiology represents the lawful manifestation of microscopic physical organization at the macroscopic level. Within TACE, this transition belongs entirely to the **Physical Level** and should not be interpreted as the emergence of consciousness itself.

Classical neuroscientific instruments, including electroencephalography (EEG), magnetoencephalography (MEG), and functional magnetic resonance imaging (fMRI), operate at spatial and temporal scales many orders of magnitude larger than the microscopic quantum organization of neuronal matter. Consequently, they cannot resolve individual quantum actualizations or the instantaneous microscopic configurations composing the neuropsychic substrate.

Instead, these instruments measure the collective physical consequences of immense populations of quantum interactions distributed throughout the nervous system. Every recorded electromagnetic oscillation represents the large-scale statistical manifestation of innumerable microscopic physical actualizations occurring simultaneously within the neuropsychic organization.

From the perspective of TACE, the instantaneous quantum organization of the brain is mathematically represented by a global tensor product state describing the current physical realization of the neuropsychic system. This mathematical description belongs entirely to the Physical Level and evolves continuously as successive lawful actualizations transform each preceding physical state ( $q-form_0$ ) into its historically enriched successor ( $q-form_1$ ).

The macroscopic electromagnetic patterns observed experimentally should therefore be understood as statistical manifestations of this underlying quantum organization rather than as direct observations of individual quantum events. The apparent continuity of electroencephalographic rhythms, cortical oscillations, and large-scale neural dynamics reflects the integration of an immense number of microscopic physical realizations whose individual discreteness remains inaccessible to classical measurement.

Accordingly, the smooth macroscopic behavior observed in neuroscience is not an illusion but a legitimate physical phenomenon arising from statistical integration across multiple scales of organization. Microscopic quantum actualizations and macroscopic electromagnetic continuity constitute complementary descriptions of the same physical reality rather than competing accounts of different realities.

Within TACE, this multi-scale organization reflects the lawful relationship between the **Matrix**, the **Q-Form**, and physical realization. The Matrix establishes the universal formal order governing admissible physical actualizations, while the Q-Form embodies the structured physical potentiality through which those actualizations become historically realized. The global tensor product state provides the mathematical representation of this evolving physical organization, whereas the macroscopic electromagnetic field represents its experimentally observable manifestation.

Nevertheless, neither the tensor product state nor the electromagnetic field constitutes consciousness. They describe the embodied neuropsychic organization through which conscious acts become historically expressed. Conscious understanding itself belongs to the activity of the substantial human soul and becomes

intentionally present through the lawful mediation of the **Silver Bridge**, which establishes the correspondence between neuropsychic realization and the Realm of Forms.

Consequently, the relationship between quantum neurodynamics and classical neuroscience should not be understood as a transition from unconscious matter to conscious experience. Rather, it represents the progressive manifestation of one and the same physical organization across different descriptive scales. Consciousness enters this framework not as an emergent property of physical complexity, but as the intentional activity of the substantial soul acting through an embodied neuropsychic organization according to the lawful ontological order established by the Creator.

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# Chapter V. Biological Perception and Multi-Scale Physical Realization

The relationship between microscopic quantum organization and macroscopic biological perception illustrates one of the fundamental principles of TACE: the same physical reality admits complementary descriptions at different scales of organization. The continuous character of ordinary sensory experience does not contradict the discrete character of quantum physical actualizations; rather, both describe the same created reality according to different levels of physical manifestation.

Within the Physical Level of TACE, microscopic interactions occur through lawful quantum actualizations governed by the Matrix and embodied through the Q-Form constraints. These physical actualizations remain individually inaccessible to classical biological perception. Instead, the neuropsychic organization integrates immense populations of microscopic physical events into coherent macroscopic sensory patterns through lawful biological processing.

Consequently, the continuity experienced in touch, smell, sight, hearing, and the remaining senses should not be interpreted as an illusion but as the legitimate macroscopic manifestation of an underlying quantum organization. Biological perception therefore represents a higher-order integration of countless microscopic physical actualizations rather than a direct observation of individual quantum events.

Within the framework of Maturana and Varela, this integrated sensory world provides the basis for operational cognition through structural coupling. Living organisms continuously learn to adapt their behavior to these embodied sensory patterns, thereby preserving their autopoietic organization and maintaining effective interaction with their environment. Their cognition is therefore fundamentally directed toward adaptive behavior.

Human beings participate in this same biological organization but transcend it through the activity of the substantial rational soul. Human consciousness does not merely respond to sensory inputs or adapt behavior to environmental conditions. Through the powers of intellect, intentionality, and will, it becomes capable of apprehending universal meanings, forming abstract concepts, judging truth and falsity, reflecting upon its own acts of knowing, and freely directing action according to rational understanding. In this way, the human person not only perceives reality but also thinks about reality, interprets its meaning, and consciously participates in its continuous historical actualization.

Accordingly, biological cognition and human consciousness are related but not identical. Biological cognition explains how living organisms successfully interact with their environment, whereas conscious understanding explains how the human person knows, evaluates, and intentionally understands the intelligible order disclosed through that interaction. The lawful correspondence between these complementary domains is established through the Silver Bridge, which mediates between embodied neuropsychic realization and the intentional presence of intelligible Forms.

Macroscopic physical systems do not interact with isolated quantum realizations individually. Rather, they interact with the instantaneous global physical configuration of surrounding systems, mathematically represented by their tensor product states or corresponding collective quantum descriptions. In living

organisms, these global physical configurations become the substrate of structural coupling, enabling adaptive biological behavior. In human beings, the same embodied neuropsychic organization becomes intentionally present to the substantial soul through the Silver Bridge, making possible not only adaptive behavior but also conceptual understanding, rational judgment, and self-reflective thought.

**Touch**

When a hand rests upon a solid surface, the experienced continuity of contact arises from the coordinated neuropsychic integration of innumerable microscopic electromagnetic interactions occurring between atoms composing the skin and the external object. No direct contact between atomic nuclei occurs. Instead, electromagnetic interactions produce a stable macroscopic pattern that the nervous system embodies as the continuous experience of solidity.

**Smell**

Olfactory perception likewise emerges from the lawful integration of discrete molecular interactions with specialized receptor systems. Individual molecular encounters become collectively organized within the neuropsychic system, producing the continuous sensory experience recognized as a persistent aroma. The continuity belongs to the integrated biological manifestation rather than to each individual molecular interaction.

**Vision**

Visual perception provides another example of multi-scale realization. Photons are absorbed individually by retinal photoreceptors according to the quantum interactions governing light. Nevertheless, the visual system integrates these discrete physical events into coherent spatial, temporal, and chromatic structures that constitute the continuous visual world ordinarily experienced.

The same principle applies throughout biological perception. Microscopic quantum actualizations remain physically discrete, while the neuropsychic organization continuously integrates these innumerable physical realizations into stable macroscopic sensory experience. Consequently, continuity and discreteness are not competing descriptions of reality but complementary manifestations of the same physical order observed at different descriptive scales.

Within TACE, this relationship reflects the lawful interaction between the Matrix, the Q-Form, and neuropsychic organization. The Matrix establishes the universal formal order governing admissible physical actualizations. The Q-Form embodies the structured physical potentiality through which these actualizations occur. Neuropsychic organization integrates the resulting physical realizations into coherent biological sensory activity.

The resulting sensory world therefore constitutes a genuine physical manifestation rather than a subjective illusion. It represents the lawful biological embodiment of countless microscopic physical actualizations organized across multiple scales of complexity.

Nevertheless, biological perception alone does not constitute conscious understanding. The neuropsychic organization provides the embodied physical conditions through which sensory information becomes available, but the intentional apprehension of meaning belongs to the activity of the substantial human soul. Through the lawful mediation of the Silver Bridge, these embodied neuropsychic realizations become intentionally present as objects of consciousness, allowing the human person not merely to perceive physical reality but also to understand, interpret, and communicate its intelligible meaning.

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# Chapter VI. Biological Cognition, Human Consciousness, and the Substantial Soul

A fundamental ontological distinction must be established between biological cognition and personal consciousness. Although every sufficiently complex living organism possesses lawful mechanisms of perception, adaptation, and environmental interaction, these biological capacities should not be identified with the intellectual consciousness proper to the human person.

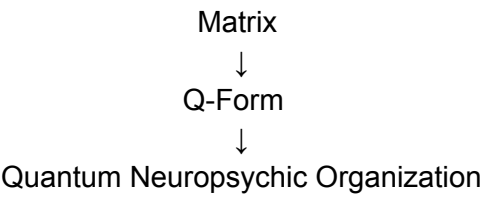
Following the enactivist framework developed by Humberto Maturana and Francisco Varela, living systems maintain themselves through autopoiesis and continuously interact with their environment through structural coupling. These principles provide an important biological account of how organisms preserve their organization, adapt to changing environments, and generate operational cognition through embodied interaction.

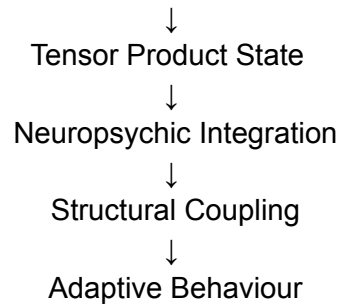
Within TACE, these biological processes belong entirely to the **Physical** and **Phenomenological–Biological Levels** of the ontological framework. They explain the organization of the neuropsychic and the lawful embodiment of sensory and motor activity, but they do not constitute a sufficient explanation of intentional understanding, universal concepts, or self-reflective intellect.

At every instant, the neuropsychic organization of both animals and human beings possesses an instantaneous global physical configuration mathematically representable as a tensor product state. This global configuration summarizes the current quantum organization of the embodied nervous system and continuously evolves through successive lawful physical actualizations governed by the Matrix and embodied in the Q-Form.

In non-rational animals, this neuropsychic organization supports perception, learning, memory, adaptation, and coordinated behavior through structural coupling with the environment. Operational cognition enables the organism to respond appropriately to changing circumstances while preserving its biological organization. Nevertheless, these processes remain directed toward adaptive interaction and do not constitute the intellectual apprehension of universal meanings.

## ANIMAL COGNITION

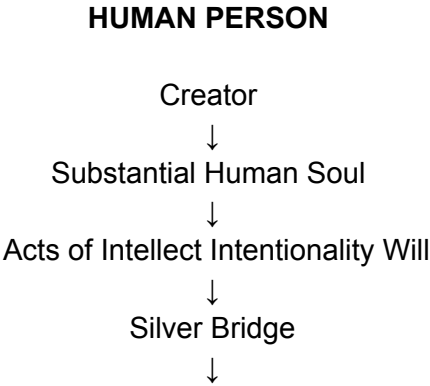




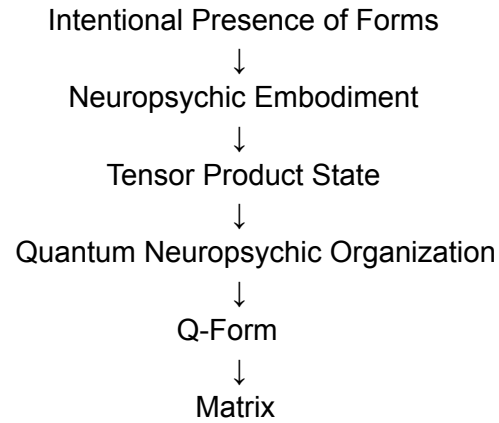
The distinctive characteristic of the human person is not merely increased biological complexity, recursive computation, or language alone. Rather, human beings possess a **substantial rational soul**, created by the Creator and constituting, together with the body, one unified human substance. The soul is the ontological principle of intellect, intentionality, and will. It is neither produced by quantum organization nor emergent from recursive biological processes.

Human language therefore occupies a privileged but secondary role. Language does not create consciousness; rather, it expresses and communicates the activity of an already existing intellect. Through language, universal meanings become historically shared, refined, preserved, and transmitted among persons. Recursive linguistic reflection enormously expands the historical development of human knowledge, yet it presupposes the existence of the rational soul capable of understanding those meanings.

The lawful correspondence between embodied neuropsychic realization and intentional understanding is mediated by the **Silver Bridge**, the universal onto-epistemological principle through which the Physical Realm and the Realm of Forms remain intrinsically interconnected. Through this mediation, neuropsychic realizations become intentionally present as intelligible objects of consciousness, while acts of intellect, intentionality, and will become lawfully embodied within the historical life of the person.







Accordingly, TACE rejects both reductive materialism and emergentist accounts that identify consciousness with increasingly complex physical organization. Likewise, it rejects substance dualism understood as the coexistence of two independently operating substances. Instead, the human person is understood as one unified substance whose embodied physical organization and substantial rational soul participate together in a single ontological order.

The tensor product state, neuropsychic organization, and biological embodiment describe the lawful physical conditions through which conscious acts become historically expressed. Conscious understanding itself, however, belongs to the substantial activity of the human soul, whose intentional acts become embodied through the lawful mediation of the Silver Bridge according to the universal formal order established by the Matrix and ultimately grounded in the creative act of the Creator.

Consequently, the distinction between animal cognition and human consciousness does not consist in the emergence of a new physical topology, nor in recursive language as the origin of consciousness. Rather, it consists in the presence of a rational substantial soul capable of universal understanding, truth, intentional freedom, and self-reflective knowledge. Language, culture, and historical development enrich and express these capacities, but they do not generate their ontological foundation.

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# How TACE Resolves the Mind–Brain Problem

For centuries, the mind–brain problem has been framed as a choice between two apparently incompatible alternatives. Materialism seeks to explain consciousness as the product of physical processes alone, whereas substance dualism separates mind and body into two independently existing realities whose interaction remains difficult to explain. Neither approach has produced a universally satisfactory account of conscious experience.

TACE proposes that this dilemma arises because the problem is formulated within an incomplete ontological framework. Instead of asking how the brain produces the mind, or how an immaterial mind interacts with matter, TACE distinguishes four integrated levels of one and the same created reality: the Semantic–Ontological, Formal–Mathematical, Physical, and Phenomenological–Biological levels.

Within this framework, the brain belongs entirely to the Physical Level. Its neuropsychic organization, quantum dynamics, and electromagnetic activity constitute the lawful physical embodiment through which conscious acts become historically expressed. The brain is therefore neither the creator of consciousness nor a passive instrument disconnected from it. Rather, it is the embodied physical realization through which the substantial human soul operates within the created order.

The human mind is not identified with the physical brain itself. It consists of the intentional activity of the substantial soul, whose powers of intellect, intentionality, and will belong to the Semantic–Ontological Level. These acts are neither reducible to quantum processes nor independent of embodied existence.

The apparent interaction problem is resolved through the Silver Bridge, the universal onto-epistemological principle mediating the lawful correspondence between the Physical Realm and the Realm of Forms. Through this mediation, neuropsychic realizations become intentionally present as intelligible objects of consciousness, while intentional acts become embodied within the historical life of the person. The relationship between mind and brain is therefore not a mysterious causal exchange between two isolated substances but the lawful coordination of complementary ontological modes belonging to one unified human substance.

The Matrix and the Q-Form further explain how physical actualizations themselves occur. The Matrix establishes the universal formal order governing admissible physical realizations, while the Q-Form embodies the structured potentiality through which physical substances become historically actualized. Consequently, the physical brain continuously evolves through lawful actualizations without requiring consciousness to produce physical reality or collapse quantum states.

The mind–brain problem therefore dissolves because its original formulation presupposes an inadequate ontology. The human person is neither a machine generating consciousness from matter nor an immaterial ghost manipulating a biological organism. Rather, the human person is one unified substance whose substantial soul and embodied neuropsychic organization participate together in the continuous historical actualization of reality.

Accordingly, consciousness is not explained by eliminating either the mind or the brain, but by assigning each its proper ontological role within the unified order of creation.



# Conclusion: Toward an Integrated Ontology of Conscious Experience

Within TACE, the Matrix embodies the Universal Formal Geometry that ontologically precedes every physical realization. Pure Geometry is therefore not generated by the Physical Realm but constitutes the universal relational order from which physical reality becomes historically actualized through the Q-Form.

Human mathematical knowledge, however, follows the opposite direction. Human beings first encounter the topology, shape, quantity, measurement, and relational organization of the Physical Realm. Through abstraction, the intellect progressively discovers numbers, geometrical relations, algebraic structures, and the broader body of mathematics. Mathematics is therefore epistemologically developed from experience while remaining ontologically grounded in the Universal Formal Geometry embodied in the Matrix.

Accordingly:

Euclidean geometry is one particular geometry.

Riemannian geometry is another.

Projective geometry, symplectic geometry, information geometry, and non-commutative geometry are particular geometric frameworks.

Hilbert spaces, tensor spaces, configuration spaces, phase spaces, and abstract ontological relational spaces are all understood as geometric organizations expressing formal relational order.

The framework developed throughout this monograph proposes that the long-standing separation between metaphysics, mathematics, physics, biology, and consciousness arises from treating complementary descriptive levels as independent realities. The Theory of Actualized Conscious Experience (TACE) argues instead that these domains constitute four integrated ontological levels of one created reality, each providing a distinct but complementary description of the same ontological order.

The **Semantic–Ontological Level** establishes the fundamental principles of being inherited from the Aristotelian-Thomistic tradition. The Creator is the necessary ground of existence, while substance, form, act, potency, soul, intellect, truth, and good constitute the primitive ontological categories through which reality is understood.

The **Formal–Structural Level** establishes the universal relational order governing every lawful actualization. Within this level, the **Matrix** constitutes the **Universal Formal Geometry** of creation: the complete relational order containing every admissible formal possibility independently of metric, dimensionality, visualization, or physical realization. Universal Formal Geometry is therefore not the geometry of spatial figures, but the universal relational architecture from which every mathematically expressible structure ultimately derives.

The **Physical Level** describes the historical realization of this formal order through quantum fields, wave functions, physical substances, and lawful actualizations. Quantum mechanics does not constitute the deepest level of reality but rather the physical description of one level of the created order. The **Q-Form** is the ontological principle through which the Universal Formal Geometry embodied in the Matrix becomes constrained into the admissible topological organizations capable of historical actualization within the Physical Realm. These admissible organizations constitute the relational framework through which quantum fields, physical substances, biological systems, and neuropsychic organization become successively realized. Physical interactions continuously transform one realized state (**q-form<sub>0</sub>**) into another (**q-form<sub>1</sub>**), thereby generating the continuous historical development of physical reality.

The **Phenomenological–Biological Level** explains how living systems organize these physical realizations through autopoiesis, structural coupling, neuropsychic integration, biological perception, and embodied cognition. Living organisms continuously adapt their behavior through structural coupling with their environment, preserving their organization by responding to the evolving physical configurations realized within the Physical Realm. Biological cognition therefore consists in adaptive participation within these embodied organizations rather than in the intentional apprehension of universal meanings.

Human beings participate fully in this biological organization while simultaneously transcending it. Human consciousness does not merely respond to sensory inputs or adapt behavior to environmental conditions. Through the substantial rational soul, the human person becomes capable of intentional understanding, universal concepts, rational judgment, truth, self-reflection, and free will. The human person therefore not only behaves within reality but also understands reality, interprets its intelligible order, and consciously participates in its continuous historical actualization.

One of the principal consequences of this ontology is the reformulation of the classical mind-brain problem. For centuries, philosophy has attempted to explain consciousness either by reducing it to physical processes or by separating mind and matter into two independently existing substances. TACE argues that both approaches arise from an incomplete ontological framework.

Within TACE, the brain belongs entirely to the Physical Level. Its neuropsychic organization provides the embodied historical realization through which conscious acts become physically expressed, but it neither produces nor constitutes consciousness itself.

Conscious understanding belongs to the substantial activity of the rational human soul, whose powers of intellect, intentionality, and will transcend purely biological organization while remaining intrinsically embodied.

The lawful correspondence between embodied neuropsychic realization and intentional understanding is mediated by the **Silver Bridge**, the universal onto-epistemological principle through which the Physical Realm and the Realm of Forms remain intrinsically interconnected. Through this lawful mediation, neuropsychic organization becomes intentionally present to the substantial soul, while acts of intellect become historically embodied within the physical life of the person.

The classical mind-brain problem is therefore not solved by discovering a new physical mechanism, nor by postulating an inexplicable interaction between independent substances. Rather, it is dissolved by recognizing that the traditional opposition between mind and matter results from an incomplete ontology. Once the Semantic–Ontological, Formal–Structural, Physical, and Phenomenological–Biological Levels are clearly distinguished and integrated, the apparent contradiction disappears. Consciousness is neither generated by matter nor detached from it. Instead, the substantial human soul acts through an embodied neuropsychic organization according to the lawful order established by the Creator through the Matrix, physically constrained by the Q-Form, and intentionally mediated by the Silver Bridge.

Accordingly, TACE neither reduces consciousness to quantum physics nor separates it from embodied existence. Quantum theory describes the lawful realization of physically admissible organizations; biology explains the organization and adaptive behavior of living systems; phenomenology describes lived conscious experience; and metaphysics establishes the ontological principles that render these complementary domains intelligible within one coherent account of reality. Each discipline preserves its own explanatory integrity while participating in one unified ontological framework.

Ultimately, the Tegmark-Aquinas Conceptual Ecosystem (TACE) is not merely a theory of consciousness, nor solely an interpretation of quantum mechanics, neuroscience, biology, or phenomenology. It is a formal ontology of created reality whose objective is to establish the lawful relationships between being, Universal Formal Geometry, physical realization, biological embodiment, intentional understanding, and conscious experience. The unity of the human person is neither an emergent property of matter nor the accidental conjunction of disconnected substances. It is the lawful participation of one created substance in four integrated ontological levels: grounded in the Creator; formally ordered through the Matrix as Universal Formal Geometry; physically constrained by the Q-Form into admissible topological organizations; embodied through neuropsychic organization; intentionally mediated by the Silver Bridge; and consciously expressed through the acts of intellect, intentionality, and will of the substantial rational soul.

The central thesis of TACE is therefore that consciousness is neither produced by matter nor separated from it. Rather, mind and brain participate together in one created ontological order whose lawful relationships become intelligible only through the integration of the Semantic–Ontological, Formal–Structural, Physical, and Phenomenological–Biological Levels. In this way, TACE proposes not merely a new theory of consciousness, but a comprehensive formal ontology of created reality in which metaphysics, Universal Formal Geometry, physics, biology, phenomenology, and theology become complementary descriptions of the lawful actualization of conscious existence.